

EXERCISE1: Control Structures

Creating Tables

```
=====
-- Create Customers Table
-- =====

CREATE TABLE Customers (
    CustomerID NUMBER PRIMARY KEY,
    CustomerName VARCHAR2(50),
    Age NUMBER,
    Balance NUMBER(10,2),
    IsVIP VARCHAR2(5)
);

-- =====
-- Create Loans Table
-- =====

CREATE TABLE Loans (
    LoanID NUMBER PRIMARY KEY,
    CustomerID NUMBER,
    InterestRate NUMBER(5,2),
    DueDate DATE,
    FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

-- =====
-- Insert Sample Data
-- =====

INSERT INTO Customers VALUES (101,'Nehal',21,12000,'FALSE');
INSERT INTO Customers VALUES (102,'Rahul',65,15000,'FALSE');
INSERT INTO Customers VALUES (103,'Priya',70,8000,'FALSE');
INSERT INTO Customers VALUES (104,'Amit',45,25000,'FALSE');
INSERT INTO Customers VALUES (105,'Sneha',62,5000,'FALSE');

INSERT INTO Loans VALUES (201,101,9.50,SYSDATE+40);
```

```

INSERT INTO Loans VALUES (202,102,10.50,SYSDATE+20);
INSERT INTO Loans VALUES (203,103,11.00,SYSDATE+15);
INSERT INTO Loans VALUES (204,104,9.00,SYSDATE+60);
INSERT INTO Loans VALUES (205,105,10.00,SYSDATE+10);

COMMIT;

```

Scenario 1:

```

-- =====
-- Scenario 1
-- Apply 1% discount to loan interest rate
-- for customers above 60 years
-- =====

BEGIN
    FOR rec IN (
        SELECT l.LoanID
        FROM Customers c
        JOIN Loans l
        ON c.CustomerID = l.CustomerID
        WHERE c.Age > 60
    )
    LOOP
        UPDATE Loans
        SET InterestRate = InterestRate - 1
        WHERE LoanID = rec.LoanID;
    END LOOP;

    COMMIT;

    DBMS_OUTPUT.PUT_LINE('Interest rates updated successfully.');
```

END;

/

```

SELECT * FROM Loans;

```

Scenario 2:

```
-- =====  
-- Scenario 2  
-- Promote customers with balance >10000  
-- to VIP  
-- =====
```

```
BEGIN  
  FOR rec IN (  
    SELECT CustomerID  
    FROM Customers  
    WHERE Balance > 10000  
  )  
  LOOP  
    UPDATE Customers  
    SET IsVIP = 'TRUE'  
    WHERE CustomerID = rec.CustomerID;  
  END LOOP;
```

```
COMMIT;
```

```
  DBMS_OUTPUT.PUT_LINE('VIP status updated successfully.');
```

```
END;  
/
```

```
SELECT * FROM Customers;
```

Scenario 3:

```
-- =====  
-- Scenario 3  
-- Print reminder for loans due within  
-- next 30 days  
-- =====
```

```
BEGIN  
  FOR rec IN (  
    SELECT c.CustomerName,  
           l.LoanID,  
           l.DueDate  
    FROM Customers c  
    JOIN Loans l  
    ON c.CustomerID = l.CustomerID  
    WHERE l.DueDate BETWEEN SYSDATE AND SYSDATE + 30  
  )
```

```

LOOP
    DBMS_OUTPUT.PUT_LINE(
        'Reminder: Dear ' || rec.CustomerName ||
        ', Loan ID ' || rec.LoanID ||
        ' is due on ' ||
        TO_CHAR(rec.DueDate, 'DD-MON-YYYY')
    );
END LOOP;
END;
/

```

Output:

Query result Script output DBMS output Explain Plan SQL history					
  Download ▼ Execution time: 0.005 seconds					
	LOANID	CUSTOMERID	INTERESTRATE	DUE DATE	
1	201	101	9.5	8/21/2026, 6:43:00	
2	202	102	9.5	8/1/2026, 6:43:00 P	
3	203	103	10	7/27/2026, 6:43:00	
4	204	104	9	9/10/2026, 6:43:00	
5	205	105	9	7/22/2026, 6:43:00	